



Python Programming - 2301CS404

Lab - 9

Roll No : 418

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01) Write a function to calculate BMI given mass and height.
($BMI = mass/h^{**2}$)

```
In [1]: def calculate_BMI(mass, height):  
        bmi = mass / (height ** 2)  
        return bmi  
  
weight = float(input("Enter your weight in kg: "))  
height = float(input("Enter your height in meters: "))  
  
result = calculate_BMI(weight, height)  
print("Your BMI is:", round(result, 2))
```

Your BMI is: 20.76

02) Write a function that add first n numbers.

```
In [3]: # def sum_first_n(n):  
#       return n * (n + 1) // 2  
  
def sum_first_n(n):  
    total = 0  
    for i in range(1, n + 1):  
        total += i  
    return total  
  
n = int(input("Enter n: "))  
result = sum_first_n(n)  
print("Sum of first", n, "numbers is:", result)
```

Sum of first 10 numbers is: 55

03) Write a function that returns 1 if the given number is Prime or 0 otherwise.

```
In [4]: def is_prime(n):
        if n <= 1:
            return 0
        for i in range(2, n):
            if n % i == 0:
                return 0

        return 1

num = int(input("Enter number: "))
print(is_prime(num))
```

1

04) Write a function that returns the list of Prime numbers between given two numbers.

```
In [6]: def primes_range(a, b):
        prime_list = []
        for num in range(a, b + 1):
            if num > 1:
                for i in range(2, num):
                    if num % i == 0:
                        break
                else:
                    prime_list.append(num)
        return prime_list

start = int(input("Enter start number: "))
end = int(input("Enter end number: "))
print(primes_range(start, end))
```

[11, 13, 17, 19]

05) Write a function that returns True if the given string is Palindrome or False otherwise.

```
In [10]: def is_palindrome(s):
        return s == s[::-1]
text = input("Enter a string: ")
print(is_palindrome(text))

# def is_palindrome(s):
#     rev = ""
#     for ch in s:
#         rev = ch + rev
#     if s == rev:
#         return True
#     else:
#         return False
# text = input("Enter a string: ")
# print(is_palindrome(text))
```

True

06) Write a function that returns the sum of all the elements of the list.

```
In [11]: # def sum_list(Lst):  
#         return sum(Lst)  
  
def sum_list(lst):  
    total = 0  
    for i in lst:  
        total += i  
    return total  
  
numbers = [1, 2, 3, 4, 5]  
print(sum_list(numbers))
```

15

07) Write a function to calculate the sum of the first element of each tuples inside the list.

```
In [12]: def sum_first_elements(tuples_list):  
    total = 0  
    for t in tuples_list:  
        total += t[0]  
    return total  
  
data = [(1, 2), (3, 4), (5, 6)]  
print(sum_first_elements(data))
```

9

08) Write a recursive function to find nth term of Fibonacci Series.

```
In [16]: def fibonacci(n):  
    if n == 0:  
        return 0  
    elif n == 1:  
        return 1  
    else:  
        return fibonacci(n-1) + fibonacci(n-2)  
  
n = int(input("Enter N: "))  
  
print("Fibonacci Series:")  
for i in range(n):  
    print(fibonacci(i), end=" ")  
  
print("\nNth Fibonacci term is:", fibonacci(n-1))
```

Fibonacci Series:

0 1 1 2 3 5 8 13 21 34

Nth Fibonacci term is: 34

09) Write a function to get the name of the student based on the given rollno.

Example: Given dict1 = {101:'Ajay', 102:'Rahul', 103:'Jay', 104:'Pooja'} find name of student whose rollno = 103

```
In [20]: def get_student_name(dict1, rollno):
        for k in dict1:
            if k == rollno:
                return dict1[k]
        return "Roll number not found"

dict1 = {101:'Ajay', 102:'Rahul', 103:'Jay', 104:'Pooja'}
roll = int(input("Enter roll number: "))
print(get_student_name(dict1, roll))
```

Jay

10) Write a function to get the sum of the scores ending with zero.

Example : scores = [200, 456, 300, 100, 234, 678]

Ans = 200 + 300 + 100 = 600

```
In [22]: # def sum_ending_zero(scores):
        #     return sum(s for s in scores if s % 10 == 0)

def sum_ending_zero(scores):
    total = 0
    for s in scores:
        if s % 10 == 0:
            total += s
    return total

scores = [200, 456, 300, 100, 234, 678]
print(sum_ending_zero(scores))
```

600

11) Write a function to invert a given Dictionary.

hint: keys to values & values to keys

Before : {'a': 10, 'b':20, 'c':30, 'd':40}

After : {10:'a', 20:'b', 30:'c', 40:'d'}

```
In [24]: # def invert_dict(d):
        #     return {v: k for k, v in d.items()}

def invert_dict(d):
    new_dict = {}
    for k, v in d.items():
        new_dict[v] = k
    return new_dict
```

```
dict1 = {'a': 10, 'b': 20, 'c': 30, 'd': 40}
print(invert_dict(dict1))
```

```
{10: 'a', 20: 'b', 30: 'c', 40: 'd'}
```

12) Write a function that returns the number of uppercase and lowercase letters in the given string.

example : Input : s1 = AbcDEfgh ,Ouptput : no_upper = 3, no_lower = 5

```
In [25]: def count_upper_lower(s):
        upper = 0
        lower = 0
        for ch in s:
            if ch.isupper():
                upper += 1
            elif ch.islower():
                lower += 1
        return upper, lower

s1 = input("Enter string: ")
u, l = count_upper_lower(s1)
print("No of Uppercase =", u)
print("No of Lowercase =", l)
```

```
No of Uppercase = 3
```

```
No of Lowercase = 5
```

13) Write a lambda function to get smallest number from the given two numbers.

```
In [26]: smallest = lambda a, b: a if a < b else b

x = int(input("Enter first number: "))
y = int(input("Enter second number: "))

print("Smallest number is:", smallest(x, y))
```

```
Smallest number is: 5
```

14) For the given list of names of students, extract the names having more that 7 characters. Use filter().

```
In [27]: students = ["Rahul", "Alexander", "Priyanka", "Jay", "Samantha", "Amitabh"]

long_names = list(filter(lambda name: len(name) > 7, students))

print(long_names)
```

```
['Alexander', 'Priyanka', 'Samantha']
```

15) For the given list of names of students, convert the first letter of all the names into uppercase. use map().

```
In [28]: students = ["rahul", "alexander", "priya", "jay", "samantha"]
```

```
capital_names = list(map(lambda name: name.capitalize(), students))

print(capital_names)
```

```
['Rahul', 'Alexander', 'Priya', 'Jay', 'Samantha']
```

16) Write udfs to call the functions with following types of arguments:

1. Positional Arguments
2. Keyword Arguments
3. Default Arguments
4. Variable Length Positional(*args) & variable length Keyword Arguments (**kwargs)
5. Keyword-Only & Positional Only Arguments

```
In [29]: # 1. Positional Arguments
def add(a, b):
    print("Sum =", a + b)

add(10, 20)
```

```
Sum = 30
```

```
In [30]: # 2. Keyword Arguments
def student(name, age):
    print("Name:", name)
    print("Age:", age)

student(age=20, name="Rahul")
```

```
Name: Rahul
```

```
Age: 20
```

```
In [31]: # 3. Default Arguments
def greet(name="Guest"):
    print("Hello", name)

greet()
greet("Ajay")
```

```
Hello Guest
```

```
Hello Ajay
```

```
In [33]: # 4. Variable Length Arguments
# (a) *args - Variable Positional
def total(*nums):
    print("Sum =", sum(nums))

total(10, 20, 30)

# (b) **kwargs - Variable Keyword
def details(**info):
    for k, v in info.items():
        print(k, ":", v)

details(name="Pooja", city="Ahmedabad", age=21)
```

```
Sum = 60  
name : Pooja  
city : Ahmedabad  
age : 21
```

In [34]: *# 5. Positional-Only & Keyword-Only Arguments*

```
def demo(a, b, /, c, d, *, e, f):  
    print(a, b, c, d, e, f)
```

```
demo(1, 2, 3, d=4, e=5, f=6)
```

1 2 3 4 5 6

In []: